Recitation - 8

21-3-26

Decision Tree classifier

ID3 algorithm.

entropy & Information Gain.

calculate IG → max to find best feature (root node)

entropy → randomness of data how data is distributed

$-\sum_{i=1}^{N} P_i * \log_2(P_i)$ → prob. of feature i

$\log_2 P_i = \frac{\log_{10} P_i}{\log_{10} 2} = \frac{\ln(P_i)}{\ln(2)}$

IG = entropy (parent)

$-\sum_{k \text{ features}} \frac{N_k}{N} \text{entropy}(k)$ features ⇒ $-\frac{1}{10}$ entropy(k)

1) entropy (parent email)

$P_{spam} = \frac{5}{10}$ → total ex $= 0.5$.

$P_{Nspam} = \frac{5}{10} = 0.5$

$\text{entropy} = -\sum_{i=1}^{2} P_i * \log_2(P_i)$

$= -0.5 * \log_2(0.5)$

$= -(-1) \Rightarrow 1.$

entropy (money keywords) = 0

they are perfectly classified so entropy = 0.

1 → spam ✓

0 → Non-spam ✗

I.G (money keywords)

= entropy (parent)

= 1 − 0 = 1.

if sender known is 1 → 5 → 1 → spam

5 → N-spam

not root node → no. of spam emails.

entropy (sender known) =

1 → spam(1)
→ N-spam(5)

$$= -\left[\frac{1}{6}\log_2\left(\frac{1}{6}\right) + \frac{5}{6}\log_2\left(\frac{5}{6}\right)\right] + \left(0 + \frac{4}{4}\log_2\left(\frac{4}{4}\right)\right)$$

entropy for 1 ⇒

0 → spam = 4

→ N-spam = 0

$$= -[-0.43 - 0.21]$$

$$= -[0.64] \Rightarrow -0.65$$

$$IG = 1 - \left(\frac{6}{10} \times 0.65 + \frac{4}{10} \times 0\right)$$

$$\Rightarrow 1 - 0.39.$$

$$= 0.61$$

not root node → IG < 1.

entropy (has attachments).

0 → spam - (2)
→ N-spam (3)
} entropy > 0.

1 → spam - 3
→ N-spam - 1
} entropy > 0

entropy (suspicious words)

entropy > 0

I.G < 1

Min entropy

MaxIG

∴ Money keywords is root node.

spam (if = 1)

(if = 0)

Non-spam → leaf nodes

depth = 1 as there is only 1 decision node

Leaf nodes = 2.